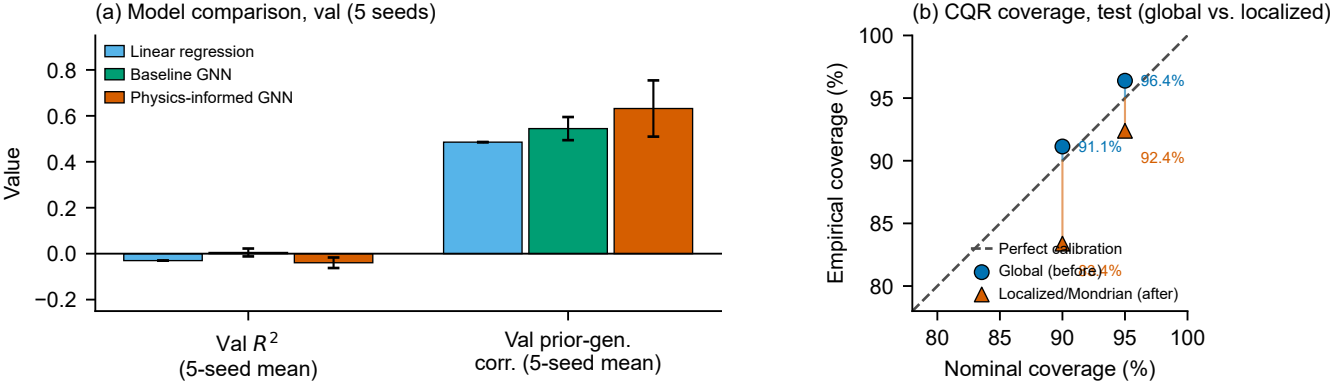
(a) Model comparison, val (5 seeds)
Linear regression
Baseline GNN
Physics-informed GNN
Value
0.8
0.6
0.4
0.2
0.0
−0.2
Val R^2
(5-seed mean)
Val prior-gen.
corr. (5-seed mean)
(b) CQR coverage, test (global vs. localized)
Empirical coverage (%)
Nominal coverage (%)
100
95
90
85
80
96.4%
91.1%
92.4%
Perfect calibration
Global (before)
Localized/Mondrian (after)